SAMPLE

Bell's Theorem

The Most Profound Discovery in Physics

Aira Astralis

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Introduction

In 1964, a thirty-four-year-old physicist at CERN, a particle theorist by day, wrote a paper in his spare time about a question the trade had declared closed, and the paper quietly ended a two-thousand-year assumption about the world. The assumption was locality: that what a thing is, and what it does, is fixed by what touches it here and now, and that influences do not jump across space in an instant. The quantum theory's entangled pairs had always looked suspicious, the far partner's answers arriving correlated as if by arrangement, and the founding generation had drawn the truce: the correlations are the theory's bookkeeping, the questions are philosophy, the physics works. John Bell asked the forbidden question, whether any account of the world that keeps the locality assumption can reproduce the quantum correlations, and he found the answer: no, and provably no, a mathematical inequality, a hard bound that any local account must obey, that the quantum predictions exceed. The experiments followed across half a century, from a basement in Berkeley to the cosmic light of distant quasars, and the verdict is in: the world is not locally made. This book tells Bell's theorem whole, in the series' plain prose, the man and his detour, the parent argument of 1935, the inequality's counting logic, the test program and its loopholes, the 2015 closures, the cosmic versions, the option space that survives, the theorem's companion, the experiments' resource era, and the honest sorting of the wilder claims, the settled physics and the labeled speculation, as this series does.

The reader who completes the tour will hold the quantum century's sharpest result in plain language: a theorem, a test, and a verdict, and the world's own nonlocality, demonstrated, replicated, and closed to argument at the labs' own finest. The book's sorting is the reader's shield: the theorem rules out one thing precisely, local hidden accounts, and everything else, the interpretations, the free-choice debates, the faster-than-light folklore, is graded carefully in the chapters ahead, the honest tour being the series' own method, from the first page to the last.

The introduction's expansion owes the reader the book's honest scope note, the series' own custom. The scope: this book being the theorem's whole story, the entanglement's physics being the neighboring volume's estate, the interpretations' full census being the measurement volume's estate, this book's holding the result itself: the logic, the tests, the verdict, the options, and the honest folklore's sorting, the reader's being able to enter here without the neighbors, the book's being self-contained in the series' plain style, and the reader's being pointed to the neighbors for the adjacent depths, the scope's note being the last of the introduction's own duties, the tour's being ready, the first chapter's being the man's own story, and the reader's walking, the whole being the honest start, the reader's first step being the next page's own, and the tour's own promise being the series' standard one: no equations, no hype, the settled and the open sorted honestly, and the world's own strangest fact made walkable, from the first sentence to the last.

Chapter 1: The Man and the Detour

The theorem's discovery chapter deserves its human record first, because the story is the series' own favorite kind: a working physicist's spare-time question. John Stewart Bell, born in Belfast in 1928, a working-class Catholic family's son, having come up through the technical colleges to a doctorate in nuclear physics and a career in the accelerators' theoretical design, joining CERN in 1960 as a particle theorist, the quantum foundations being officially a hobby, the hobby having been fed by his own decade of private doubts, the accelerator world's by-day business being the symmetries and the particle dynamics, and the foundations being the by-night reading. The trigger: Bohm's 1952 pilot-wave papers, the deterministic reconstruction of quantum mechanics that the trade had mostly ignored, Bell having been assigned to review it for a seminar and having come away impressed and bothered, the bothered part being the question: how does Bohm's explicit nonlocality relate to the standard theory's apparent locality, and can a hidden account be built that keeps the locality? The question sat in the drawer, and the sabbatical of 1963-64, a year in the United States, at Stanford and Wisconsin and Brandeis, being the theorem's own season, the paper written and submitted in 1964, the famous result being a hobby's harvest, the trade's own reception being silence for years, the paper having been published in a short-lived journal, Physics, that most libraries did not hold, the obscurity being the result's first decade, the theorem having been discovered in effect twice, once in 1964 and again in the 1970s when the experimenters found it.

Bell's own personality deserves the honest portrait: the Irishman's bluntness being the trade's own legend, the seminars' sharpest questioner, the foundations' most uncompromising voice, the papers' prose being famously careful, the 1964 paper's title being modest, the theorem having been named by others. The reader should hold the portrait's honest meaning: the result came from the trade's own inside, a fully credentialed particle physicist, not a marginal figure, and the result's obscurity being the culture's own doing, the truce having been so complete that a revolution could sit in an unread journal for a decade, the story being the series' recurring one: the question the truce refused having returned, by way of a hobby, wearing a CERN badge.

The discovery's second visit is the sabbatical's own record, because the theorem's birth season deserves the honest detail. The 1963-64 leave having been taken by Bell and his wife Mary, herself a physicist and accelerator colleague, the two having toured the American stations, the Stanford term being the paper's own writing room, and the Wisconsin and Brandeis visits being the argument's own stages, the seminars' receptions being mixed, the trade's foundations allergy being the era's own medicine, the paper having been submitted from the leave's travels, the season's honest meaning being the result's fragility: had the sabbatical gone otherwise, had the drawer question stayed drawn, the trade's decade of silence having been nearly a death, the theorem's survival owing to a few experimenters' stubbornness in the 1970s, the season being the reminder that the science's key results often hang by single threads of biography, the reader's honest awe being the fragility itself.

Bell's own later standing deserves the record, the man's having lived to see the turn: the 1980s' CERN seminar rooms having hosted the foundations' own revival, Bell being the trade's conscience in person, the workshops being organized, the Erice meetings being the foundations' own parliament, Bell's being the movement's presiding skeptic, the questions' having been sharpened by his presence, and the man's death in 1990 having cut the revival's own season short, the theorem's 2015 closure being thus posthumous, the verdict having arrived without its judge, the honest record being that Bell died before the loophole-free tests, the question's having been settled by the heirs, the man's own papers being the heirs' own instruments, the biography's closing note being the series' recurring one: the question's askers rarely see the answer's finish line, the asker's monument being the question itself.

Bell's own writings deserve the second record, because the man's essays are the foundations' best prose. The collected works, the two volumes of Speakable and Unspeakable, the title itself being the locality question's own joke, the speakable being the observable physics, the unspeakable being the hidden account, the essays' being the theorem's own companion literature, the Bertlmann socks piece being the collection's most quoted, the introductions' being the program's own honest running commentary, the reader's being pointed to the essays as the book's own best companion, the man's prose being the series' own standard: exact, blunt, and generous to the opponents, Bell's having written the foundations' plain language before this series tried, the essays being the inheritance's own shelf.

The man's third record is the Belfast roots, because the biography's beginnings deserve their lines. The Protestant-dominated city's Catholic family, the farm family's son, the scholarship's being the ladder, the technical college's being the first rung, the Queen's University Belfast being the second, the accelerator physics being the career's road, the road's having been the era's own narrow path for the working-class scholar, Bell's own recollections being the record of a talent requiring the era's rare breaks, the biography's honest meaning being the sociology's: the century's physics having been built by the lucky survivors of many filters, the theorem's author being one, the reader's holding the roots as the series' honest sociology, the luck's being recorded without romance, the breaks' being named, and the talent's having been the necessary and insufficient condition, the drawer question's having needed the career's badge to be asked safely, the badge's having needed the ladder's luck to exist at all.